Chapter 4

Potentials in Position-Dependent Mass Background and Exceptional Orthogonal Polynomials

In the previous chapter, we studied the EOPs within the framework of SUSY. In this chapter, we obtain some ES potentials in the PDM framework and the associated eigenfunctions related to X_m-Laguerre EOPs. First, we use the PCT approach in the PDM framework to construct ES potentials. Due to its importance in describing the physics of many current microstructures, PDM has attracted notable attention. The mathematics used in the PCT approach is described in Chapter 2. We use two scenarios for PDM. In the first case, we take a form of the PDM that has already been used in previous works and construct ES potentials. The eigenfunctions corresponding to these potentials are written in terms of X_m-Laguerre polynomials. We show that for a particular value, $m = 1$, the eigenvalues and eigenfunctions are matched with the Midya et al. case [231]. In the second case, we introduced a new form of the PDM that depends on a parameter ν [232]. In this case, we obtain the potentials, coresponding eigenvalues, and eigenfunctions. Furthermore, using the SUSY technique in the PDM framework, we obtain the superpotentials and corresponding partner potentials for both cases. We find that in both cases, one of the potentials exactly matches the previously obtained potentials. This indicates that the potentials are shape invariant, which is the key reason for their exact solvability.

4.1 Schrödinger Equation in PDM framework

To address the issue of noncommutativity, we begin with the von Roos prescription for the two-parameter formulation of an effective-mass kinetic energy operator, as described in [248]. This approach inherently ensures Hermiticity and allows for various plausible special cases. The operator is expressed as follows

$$\hat{T} = \frac{1}{4}[\mathcal{M}^r(x)p\mathcal{M}^s(x)p\mathcal{M}^t(x) + \mathcal{M}^t(x)p\mathcal{M}^s(x)p\mathcal{M}^r(x)]. \tag{4.1}$$

where r, s and t are uncertain parameters that satisfy the condition $r+s+t=-1$. Using the kinetic energy operator $\hat{T}$ mentioned above, the Hamiltonian for PDM Schrödinger can be written as

$$H = \Big[\frac{1}{4}[\mathcal{M}^r(x)p\mathcal{M}^s(x)p\mathcal{M}^t(x) + \mathcal{M}^t(x)p\mathcal{M}^s(x)p\mathcal{M}^r(x)] + V(x)\Big], \tag{4.2}$$

here, $V(x)$ represents the external potential. Several common Hermitian Hamiltonians have been examined in the literature, with different choices of the uncertain parameters. For further details, refer to [249] and the references therein.

By adopting units where $\hbar = 2\mathcal{M}_0 = 1$, the PDM Schrödinger corresponding to the Hamiltonian in (4.2) can be expressed as

$$\Big[-\frac{1}{2}\Big(M^r(x)\frac{d}{dx}M^s(x)\frac{d}{dx}M^t(x) + M^t(x)\frac{d}{dx}M^s(x)\frac{d}{dx}M^r(x)\Big) + V(x)\Big]\psi(x) = E\psi(x). \tag{4.3}$$

Here, $M(x)$ is defined as $\mathcal{M}(x) = \mathcal{M}_0 M(x)$ and is dimensionless. By setting $M(x) = f^{-2}(x)$ and $f(x) = 1 + \mathfrak{g}(x)$, as described in [245], where $f(x)$ is a positive-definite function and $\mathfrak{g}(x) = 0$ represents the constant-mass case, Eq. (4.3) transforms into

$$\Big[-\frac{1}{2}\Big(f^r(x)\frac{d}{dx}f^s(x)\frac{d}{dx}f^t(x) + f^t(x)\frac{d}{dx}f^s(x)\frac{d}{dx}f^r(x)\Big) + V(x)\Big]\psi(x) = E\psi(x), \tag{4.4}$$

with $r+s+t=2$. The uncertain parameters r,s,t (collectively denoted by r) in the kinetic energy term can be eliminated by incorporating them into the effective potential energy of the variable-mass system. Thus, by utilizing the outcome

$$\begin{aligned} f^r\frac{d}{dx}f^s\frac{d}{dx}f^t + f^t\frac{d}{dx}f^s\frac{d}{dx}f^r = {} & 2\sqrt{f}\frac{d}{dx}f\frac{d}{dx}\sqrt{f} - (1-r-t)ff'' \\ & - 2(\frac{1}{2}-r)(\frac{1}{2}-t)f'^2, \end{aligned} \tag{4.5}$$

where prime indicate the differentiation with respect to x and assuming the positive definiteness of f, Eq. (4.4) takes the following form

$$H\psi(x) = \Big[-\Big(\sqrt{f(x)}\frac{d}{dx}\sqrt{f(x)}\Big)^2 + V_{eff}(x)\Big]\psi(x) = E\psi(x) \tag{4.6}$$

resulting the effective potential

$$V_{eff}(x) = V(x) + \frac{1}{2}(1-r-t)f(x)f''(x) + (\frac{1}{2}-r)(\frac{1}{2}-t)f'^2(x). \tag{4.7}$$

The PDMSE (4.6) can be understood as the deformed Schrödinger equation, which is expressed as

$$H\psi(x) = \Big[\pi^2 + V_{eff}(x)\Big]\psi(x) = E\psi(x), \tag{4.8}$$

here the standard momentum operator $p = -i\frac{d}{dx}$ is changed by the deformed quantity $\pi = -i\sqrt{f(x)}\frac{d}{dx}\sqrt{f(x)}$. As a result, the conventional commutation relation $[x, p] = i\hbar$ is modified to $[x, \pi] = i\hbar f(x)$, where $f(x)$ acts as a deforming function [245, 250, 251]. This form of deformation has been extensively investigated in various fields of physics, including statistical physics (referred to as $q-$deformed algebra) [252, 253], optics [254, 255], and black hole physics [256, 257]. In the next section, we are going to use the PCT and EOPs to produce a set of solvable potentials and their corresponding eigen-spectrum.

4.2 Generation of new potential by PCT approach

By using the technique described in the preceding section, the PDMSE in Eq. (4.3) can be expressed as

$$H\psi(x) = \Big[-\frac{d}{dx}\frac{1}{M(x)}\frac{d}{dx} + V_{eff}\Big]\psi(x) = E\psi(x), \tag{4.9}$$

where the effective potential is given by

$$V_{eff}(x) = V(x) + \frac{s+1}{2}\frac{M''}{M^2} - [r(r+s+1)+s+1]\frac{M'^2}{M^3}. \tag{4.10}$$

The term $V(x)$ represents the potential applied in the system. We shall adopt the selection made by BenDaniel and Duke [258], namely setting $r = t = 0$ and $s = -1$. The choice made leads to the effective potentials becoming equal to the applied potentials. We adopt the solution $\psi(x)$ as described in the form provided by [231, 244, 245]

$$\psi(x) = f(x)F(g(x)), \tag{4.11}$$

where $F(g(x))$ follows a second-order differential equation

$$\frac{d^2F}{dg^2} + Q(g)\frac{dF}{dg} + R(g)F = 0. \tag{4.12}$$

Now, by substituting $\psi(x)$ in Eq. (4.9), we obtain

$$\frac{d^2F}{dg^2} + \left(\frac{g''}{g'^2} + \frac{2f'}{fg'} - \frac{M'}{Mg'}\right)\frac{dF}{dg} + \Big(\frac{f''}{fg'^2} - \frac{M'f'}{Mfg'^2} + (E - V_{eff})\frac{M}{g'^2}\Big)F = 0. \tag{4.13}$$

After performing a comparison, we obtain

$$Q(g(x)) = \Big(\frac{g''}{g'^2} + \frac{2f'}{fg'} - \frac{M'}{Mg'}\Big), \tag{4.14}$$

$$R(g(x)) = \left(\frac{f''}{fg'^2} - \frac{M'f'}{Mfg'^2} + (E - V_{eff})\frac{M}{g'^2}\right). \tag{4.15}$$

Eq. (4.14) has a solution as stated in [244].

$$f(x) \propto \sqrt{\frac{M}{g'}}e^{\frac{1}{2}\int^{g(x)} Q(t)dt}. \tag{4.16}$$

Using Eqs.(4.16) and (4.15), we obtain

$$E - V_{eff} = \frac{g'''}{2Mg'} - \frac{3}{4M}\left(\frac{g''}{g'}\right)^2 + \frac{g'^2}{M}\left(R - \frac{1}{2}\frac{dQ}{dg} - \frac{Q^2}{4}\right) - \frac{M''}{2M^2} + \frac{3M'^2}{4M^3}. \tag{4.17}$$

In order to balance the term E on the left-hand side of the equation, we may incorporate an appropriate relationship between $M(x)$ and $g(x)$, which will provide an energy-like component on the right-hand side. This leads to the formation of an effective potential $V_{eff}(x)$, and the solution to the related Schrödinger equation will be well-behaved. Three distinct cases for $M(x)$, where $M(x) \propto (g'(x))^{\mu}$ with $\mu = -1, 1, 2$, have been commonly used in various applications of the PCT technique within the PDM framework [259, 260]. Here, we consider a new type of PDM

$$M(x) = c\left(g'(x),\right)^{\nu} \tag{4.18}$$

where c is denoted as proportionality constant, $\nu = \frac{2\eta}{2\eta+1}$ and $\eta = 0, 1, 2\cdots$ along with a well-known PDM, $M(x) = \lambda g'(x)$.

4.2.1 Case 1

Choosing $M(x) = \lambda g'(x)$, Eq. (4.17) takes on a new form,

$$E - V_{eff} = \frac{g'}{\lambda}\Big(R - \frac{1}{2}\frac{dQ}{dg} - \frac{Q^2}{4}\Big). \tag{4.19}$$

After that, we compare Eq. (4.12) with the differential equation for X_m-Laguerre polynomials to obtain

$$\begin{aligned} Q &= \frac{1}{g}\left((\alpha + 1 - g) - 2g\frac{L^{\alpha}_{m-1}(-g)}{L^{\alpha-1}_{m}(-g)}\right), \quad \text{and} \\ R &= \frac{1}{g}\left(n - 2\alpha\frac{L^{\alpha}_{m-1}(-g)}{L^{\alpha-1}_{m}(-g)}\right). \end{aligned} \tag{4.20}$$

here $F(g) \propto \hat{L}^{(\alpha)}_{n,m}(g)$.
For these values of $Q(g(x))$ and $R(g(x))$ Eq. (4.19) can be written as,

$$\begin{aligned} E - V_{eff} = &\frac{g'}{\lambda g}\left(n - \left(m - \frac{1}{2}\right) + \frac{\alpha}{2} + \frac{m}{\alpha}\right) - \frac{g'}{4\lambda} - \frac{g'}{4g^2\lambda}\left(\alpha^2 - 1\right) \\ &- \frac{2g' L^{\alpha}_{m}(-g)^2}{\lambda\left(L^{\alpha-1}_{m}(-g)\right)^2} + \frac{g' L^{\alpha+1}_{m-2}(-g)}{\lambda\left(L^{\alpha-1}_{m}(-g)\right)} - \frac{g' L^{\alpha}_{m-1}(-g)}{\lambda\left(L^{\alpha-1}_{m}(-g)\right)} \\ &- \frac{g' L^{\alpha+1}_{m-1}(-g)}{\alpha\lambda\left(L^{\alpha-1}_{m}(-g)\right)} + \frac{g' L^{\alpha+1}_{m-1}(-g)}{\lambda\left(L^{\alpha-1}_{m}(-g)\right)}. \end{aligned} \tag{4.21}$$

It is observed that by choosing $\frac{g'}{\lambda g} = C$, the energy term on the left side will be equal to a constant term on the right side of the equation. To ensure increasing eigenvalues for successive n-values, C must be positive. Solving the equation $\frac{g'}{\lambda g} = C$ leads to

$$g(x) = e^{-bx} \text{ and } M(x) = e^{-bx}, \quad -\infty < x < \infty \tag{4.22}$$

here we have considered $C = b^2$ and $C\lambda = -b,\ b > 0$.
By substituting Eq. (4.22) into Eq. (4.21), we obtain the effective potential and energy eigenvalues as follows

$$V_{eff} = \frac{b^2}{4}\left(e^{bx}\left(\alpha^2-1\right)+e^{-bx}\right)+b^2\left(\frac{2(L^{\alpha}_{m-1}(-e^{-bx}))^2}{\left(L^{\alpha-1}_{m}(-e^{-bx})\right)^2}-\frac{L^{\alpha+1}_{m-2}(-e^{-bx})}{L^{\alpha-1}_{m}(-e^{-bx})}\right.$$
$$\left.+\frac{L^{\alpha+1}_{m-1}(-e^{-bx})}{\alpha L^{\alpha-1}_{m}(-e^{-bx})}+\frac{L^{\alpha}_{m-1}(-e^{-bx})}{L^{\alpha-1}_{m}(-e^{-bx})}-\frac{L^{\alpha+1}_{m-1}(-e^{-bx})}{L^{\alpha-1}_{m}(-e^{-bx})}\right)e^{-bx}+V^I_c, \tag{4.23}$$

$$E^m_n = b^2\left(n+\frac{\alpha+1}{2}+\frac{m}{\alpha}\right)+V^I_c, \quad n=0,1,2,... \tag{4.24}$$

with an arbitrary constant V^I_c. The eigenfunctions are obtained using Eqs. (4.11) and (4.16)

$$\psi^m_n(x) = \mathcal{N}\frac{\exp\left[-\frac{1}{2}((\alpha+1)bx+e^{-bx})\right]}{L^{\alpha-1}_{m}(-e^{-bx})}\hat{L}^{\alpha}_{n+m,m}(e^{-bx}) \quad n=0,1,2,... \tag{4.25}$$

$\mathcal{N}$ is the normalization constant given by

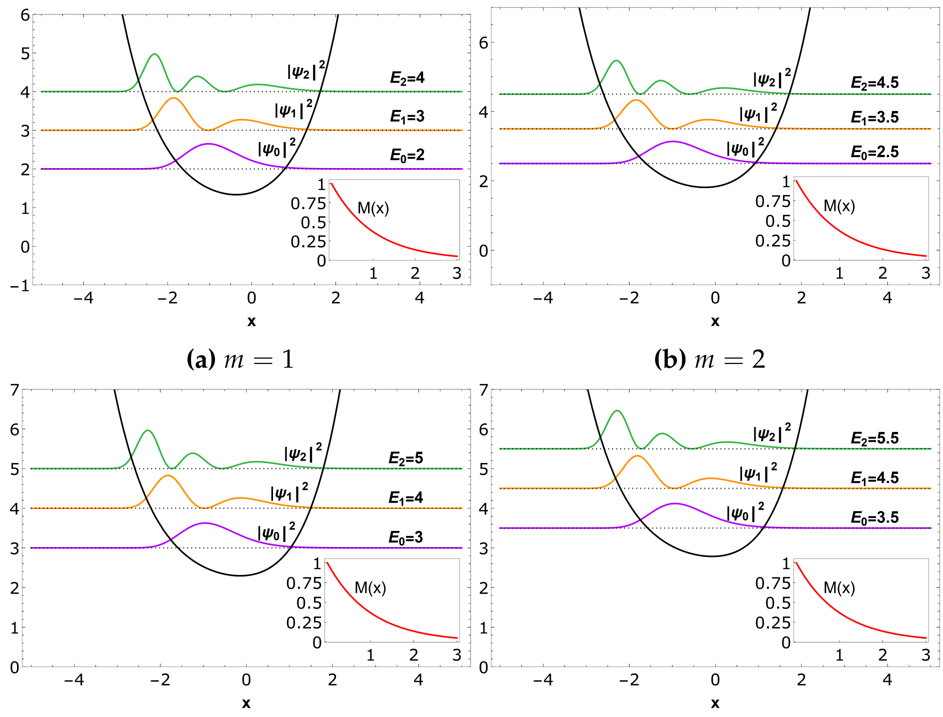
$$\mathcal{N} = \left(\frac{b\, n!}{(n+m+\alpha)\Gamma(n+\alpha)}\right)^{\frac{1}{2}}.$$

It is worth noting that the case for $m=1$ was addressed in [231]. Our general result for the X_m case replicates the findings of [231] as follows

$$V_{eff} = \frac{b^2}{4}\left(e^{bx}(\alpha^2-1)+e^{-bx}+\frac{4}{\alpha(1+\alpha e^{bx})}+\frac{8}{(1+\alpha e^{bx})^2}\right)+V^I_c, \tag{4.26}$$

$$E^1_n = b^2\left(n+\frac{\alpha+1}{2}+\frac{1}{\alpha}\right)+V^I_c, \quad n=0,1,2,... \tag{4.27}$$

$$\psi^1_n(x) = \mathcal{N}\frac{\exp\left[-\frac{1}{2}((\alpha+1)bx+e^{-bx})\right]}{\alpha+e^{-bx}}\hat{L}^{\alpha}_{n+1}(e^{-bx}) \quad n=0,1,2,... \tag{4.28}$$



(a) $m = 1$ **(b)** $m = 2$ **(c)** $m = 3$ **(d)** $m = 4$

Figure 4.1: Plot of the potential V_{eff} given in Eq. (4.23) for different m values, square of first three bound state wavefunctions $|\psi_0|^2$ (purple line), $|\psi_1|^2$ (orange line), $|\psi_2|^2$ (green line) corresponding to different m values, for mass function $M(x)$ (red line) given in Eq. (4.22). We have considered here $b = 1$, $\alpha = 2$.

4.2.2 Case 2

In this case, we introduce a new type of PDM given by $M(x) = c\,(g'(x))^{\nu}$, where $\nu = \frac{2\eta}{2\eta+1}$ and $\eta = 0, 1, 2, \ldots$. With this choice, Eq. (4.17) becomes

$$E - V_{eff} = \frac{g'''}{2cg'^{(\nu+1)}} - \frac{3g''^2}{4cg'^{(\nu+2)}} + \frac{1}{cg'^{(\nu-2)}}\left(R - \frac{1}{2}\frac{dQ}{dg} - \frac{Q^2}{4}\right) - \frac{(\nu^2-\nu)g''^2}{2cg'^{(\nu+2)}} - \frac{\nu g'''}{2cg'^{(\nu+1)}} + \frac{3\nu^2 g''^2}{4cg'^{(\nu+2)}}. \tag{4.29}$$

Next, we consider the differential equation for X_m-Laguerre polynomials using Eq. (4.12) to obtain

$$Q = \frac{1}{g}\left((\alpha+1-g) - 2g\frac{L^{\alpha}_{m-1}(-g)}{L^{\alpha-1}_{m}(-g)}\right), \quad \text{and} \quad R = \frac{1}{g}\left(n - 2\alpha\frac{L^{\alpha}_{m-1}(-g)}{L^{\alpha-1}_{m}(-g)}\right). \tag{4.30}$$

where, $F(g) \propto \hat{L}^{(\alpha)}_{n,m}(g)$.
With these values of $Q(g(x))$ and $R(g(x))$, Eq. (4.29) becomes

$$\begin{aligned}
E - V_{eff} = & \frac{g'^{(2-\nu)}}{gc}\left(n - \left(m - \frac{1}{2}\right) + \frac{\alpha}{2} + \frac{m}{\alpha}\right) - \frac{g'^{(2-\nu)}}{4c} - \frac{g'^{(2-\nu)}}{4g^2 c}\left(\alpha^2 - 1\right) \\
& - \frac{2g'^{(2-\nu)} L^{\alpha}_{m}(-g)^2}{c\left(L^{\alpha-1}_{m}(-g)\right)^2} + \frac{g'^{(2-\nu)} L^{\alpha+1}_{m-2}(-g)}{c\left(L^{\alpha-1}_{m}(-g)\right)} - \frac{g'^{(2-\nu)} L^{\alpha}_{m-1}(-g)}{c\left(L^{\alpha-1}_{m}(-g)\right)} \\
& - \frac{g'^{(2-\nu)} L^{\alpha+1}_{m-1}(-g)}{\alpha c\left(L^{\alpha-1}_{m}(-g)\right)} + \frac{g'^{(2-\nu)} L^{\alpha+1}_{m-1}(-g)}{c\left(L^{\alpha-1}_{m}(-g)\right)} - \frac{(\nu^2 - \nu) g''^2}{2cg'^{(\nu+2)}} \\
& - \frac{\nu g'''}{2cg'^{(\nu+1)}} + \frac{3\nu^2 g''^2}{4cg'^{(\nu+2)}} + \frac{g'''}{2cg'^{(\nu+1)}} - \frac{3g''^2}{4cg'^{(\nu+2)}}.
\end{aligned} \tag{4.31}$$

We find that choosing $g'^{(2-\nu)} = \kappa\, c\, g$, where κ is a constant, yields a constant term on the right-hand side of the equation, matching the energy term E on the left-hand side. To ensure increasing eigenvalues for successive n values, the constant κ must be restricted to positive values. The solution to the first-order differential equation $g'^{(2-\nu)} = \kappa\, c\, g$ is $g(x) = x^{\frac{2-\nu}{1-\nu}}$, which gives $M(x) = \left(\frac{2-\nu}{1-\nu}\right)^2 x^{\frac{\nu}{1-\nu}}$. For simplicity, we can choose $\kappa\, c = \left(\frac{2-\nu}{1-\nu}\right)^2$, setting $\kappa = 1$ and $c = \left(\frac{2-\nu}{1-\nu}\right)^2$. By substituting these values into Eq. (4.31), we obtain the effective potential and energy eigenvalues as

$$\begin{aligned}
V_{eff} = & \frac{1}{4}\left(x^{\frac{\nu-2}{1-\nu}}\left(\alpha^2 - 1\right) + x^{\frac{2-\nu}{1-\nu}}\right) + \left(\frac{2(L^{\alpha}_{m-1}(-x^{\frac{2-\nu}{1-\nu}}))^2}{(L^{\alpha-1}_{m}(-x^{\frac{2-\nu}{1-\nu}}))^2} - \frac{L^{\alpha+1}_{m-2}(-x^{\frac{2-\nu}{1-\nu}})}{L^{\alpha-1}_{m}(-x^{\frac{2-\nu}{1-\nu}})}\right. \\
& \left. + \frac{L^{\alpha+1}_{m-1}(-x^{\frac{2-\nu}{1-\nu}})}{\alpha L^{\alpha-1}_{m}(-x^{\frac{2-\nu}{1-\nu}})} + \frac{L^{\alpha}_{m-1}(-x^{\frac{2-\nu}{1-\nu}})}{L^{\alpha-1}_{m}(-x^{\frac{2-\nu}{1-\nu}})} - \frac{L^{\alpha+1}_{m-1}(-x^{\frac{2-\nu}{1-\nu}})}{L^{\alpha-1}_{m}(-x^{\frac{2-\nu}{1-\nu}})}\right) x^{\frac{2-\nu}{1-\nu}} \\
& + \frac{(1-\nu)(3-\nu)}{4(2-\nu)^2} x^{\frac{\nu-2}{1-\nu}} + V^{II}_{c},
\end{aligned} \tag{4.32}$$

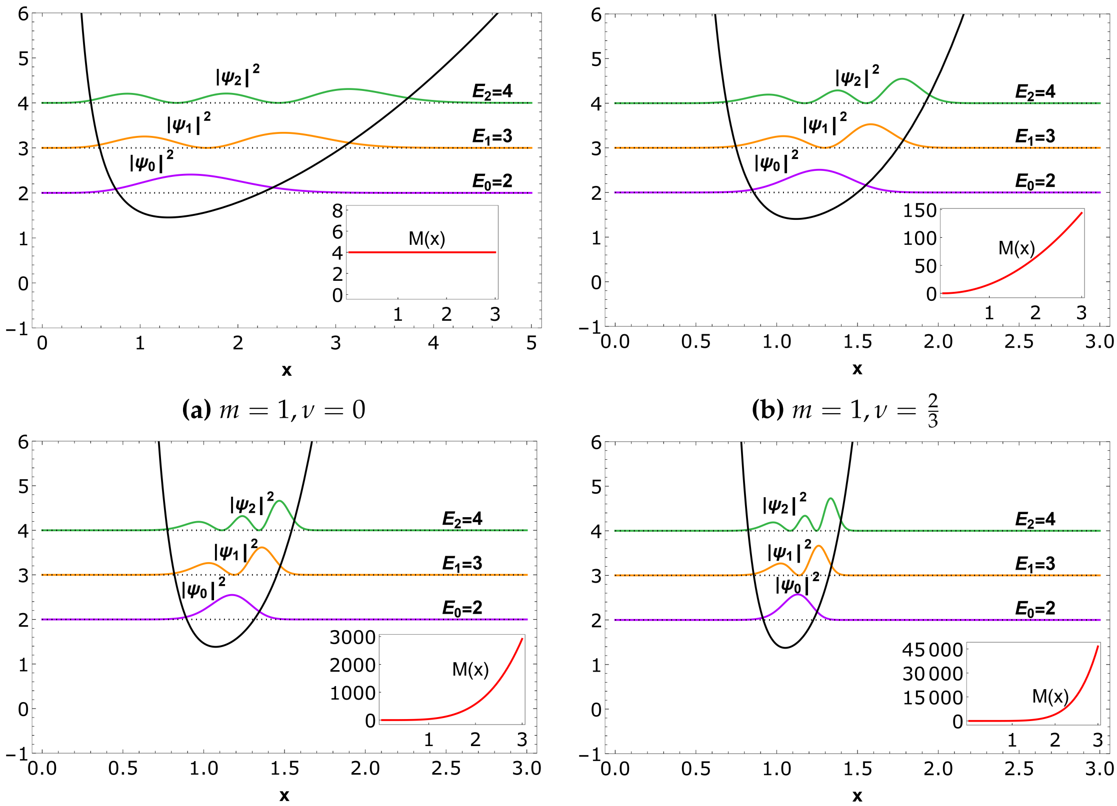
$$E^{m}_{n} = \left(n + \frac{\alpha+1}{2} + \frac{m}{\alpha}\right) + V^{II}_{c}, \quad n = 0, 1, 2, ... \tag{4.33}$$

with V^{II}_{c} as an arbitrary constant. The eigenfunctions are obtained using Eqs. (4.11) and (4.16)

$$\psi^{m}_{n}(x) = \mathcal{N} \frac{\exp\left(\frac{-x^{\frac{2-\nu}{1-\nu}}}{2}\right)}{x^{\frac{1}{2}}\left(L^{\alpha-1}_{m}(-x^{\frac{2-\nu}{1-\nu}})\right)} x^{\frac{(2-\nu)(1+\alpha)}{2(1-\nu)}} \hat{L}^{\alpha}_{n+m,m}(x^{\frac{2-\nu}{1-\nu}}), \quad n = 0, 1, 2, ... \tag{4.34}$$

here $\mathcal{N}$ is the normalization constant and it is given by

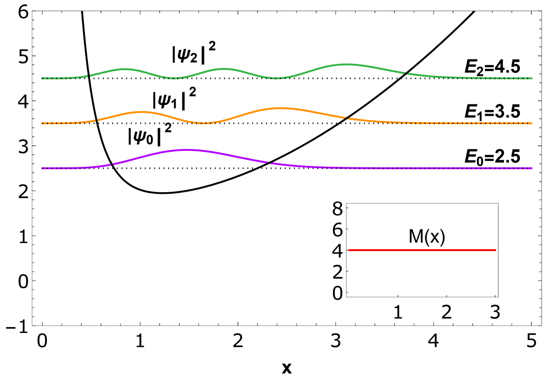
$$\mathcal{N} = \left(\frac{n!}{(n+m+\alpha)\Gamma(n+\alpha)} \right)^{\frac{1}{2}}.$$



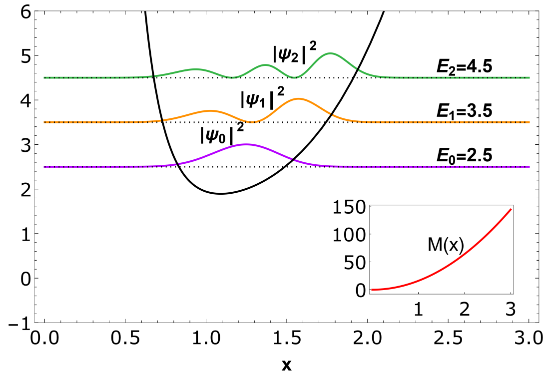
(a) $m = 1, \nu = 0$ **(b)** $m = 1, \nu = \frac{2}{3}$

(c) $m = 1, \nu = \frac{4}{5}$ **(d)** $m = 1, \nu = \frac{6}{7}$

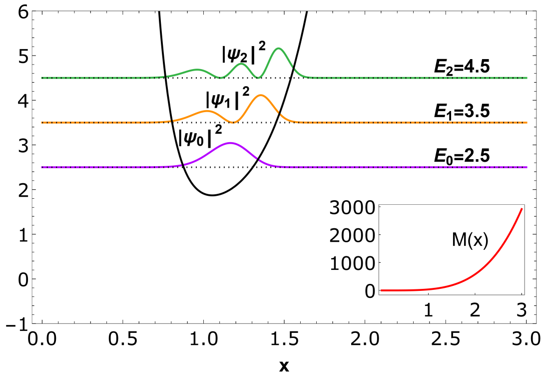
Figure 4.2: Plot of the potential V_{eff} given in Eq. (4.32) for $m = 1$ and different ν values, square of first three bound state wavefunctions $|\psi_0|^2$ (purple line), $|\psi_1|^2$ (orange line), $|\psi_2|^2$ (green line) corresponding to different ν values, for mass function $M(x)$ (red line). We have considered here $\alpha = 2$.



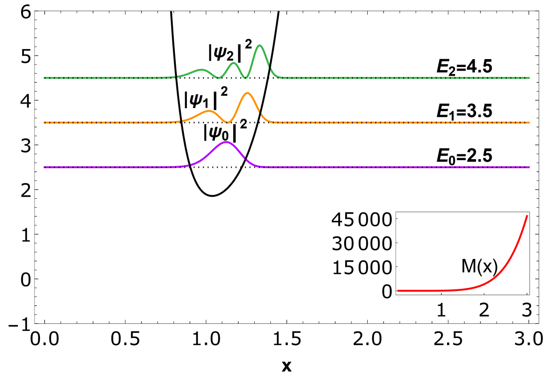
(a) $m = 2, \nu = 0$



(b) $m = 2, \nu = \frac{2}{3}$

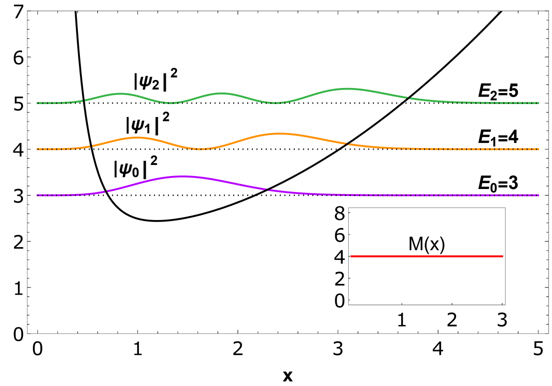


(c) $m = 2, \nu = \frac{4}{5}$

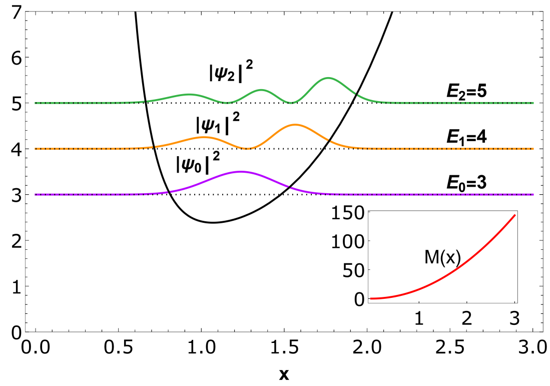


(d) $m = 2, \nu = \frac{6}{7}$

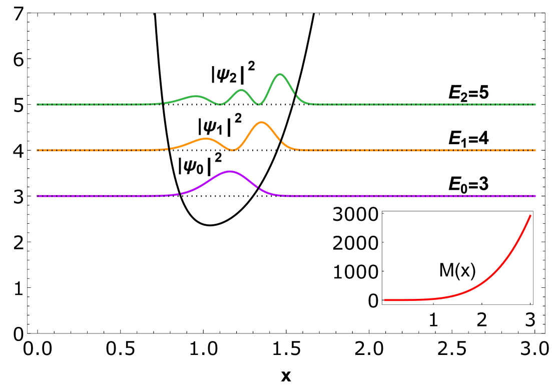
Figure 4.3: Plot of the potential V_{eff} given in Eq. (4.32) for $m = 2$ and different ν values, square of first three bound state wavefunctions $|\psi_0|^2$ (purple line), $|\psi_1|^2$ (orange line), $|\psi_2|^2$ (green line) corresponding to different ν values, for mass function $M(x)$ (red line). We have considered here $\alpha = 2$.



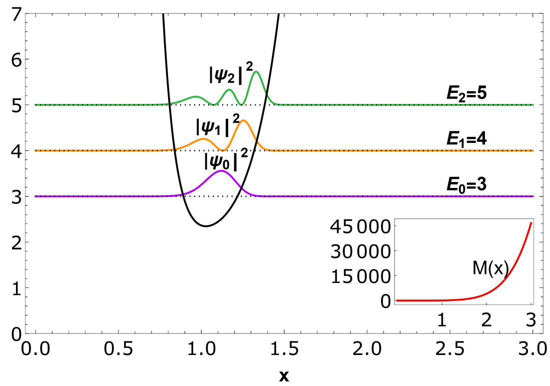
(a) $m = 3, \nu = 0$



(b) $m = 3, \nu = \frac{2}{3}$



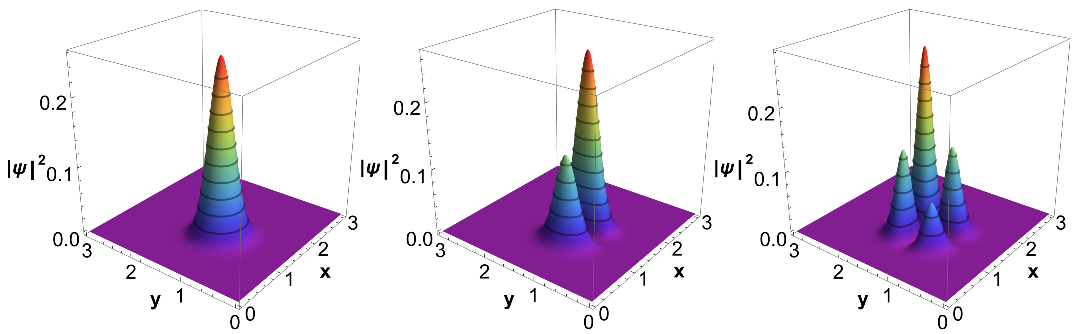
(c) $m = 3, \nu = \frac{4}{5}$



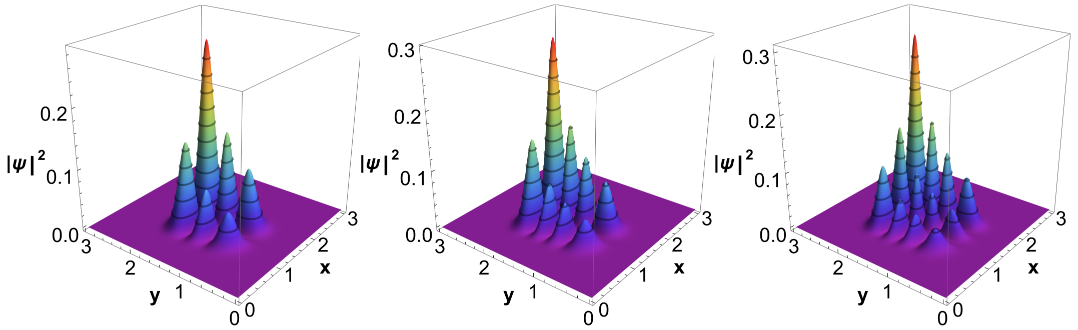
(d) $m = 3, \nu = \frac{6}{7}$

Figure 4.4: Plot of the potential V_{eff} given in Eq. (4.32) for $m = 3$ and different ν values, square of first three bound state wavefunctions $|\psi_0|^2$ (purple line), $|\psi_1|^2$ (orange line), $|\psi_2|^2$ (green line) corresponding to different ν values, for mass function $M(x)$ (red line). We have considered here $\alpha = 2$.

We further extend this analysis to the two-dimensional case. The probability density of a two-dimensional potential is shown in the Fig. 4.5.



(a) $n_1 = 0, n_2 = 0$ **(b)** $n_1 = 1, n_2 = 0$ **(c)** $n_1 = 1, n_2 = 1$



(d) $n_1 = 1, n_2 = 2$ **(e)** $n_1 = 1, n_2 = 3$ **(f)** $n_1 = 2, n_2 = 3$

Figure 4.5: Plot for the probability density of a two-dimensional potential for $m = 1, \nu = \frac{2}{3}$ and different combinations of n_1 and n_2 values. The wavefunctions are normalized.

4.3 Supersymmetric approach

In the context of SUSY QM in PDM framework, one can define the lowering and raising operators [86]

$$\mathcal{A}\psi = \frac{1}{\sqrt{M}}\frac{d\psi}{dx} + \mathcal{W}\psi, \quad \text{and} \quad \mathcal{A}^{\dagger}\psi = -\frac{d}{dx}\left(\frac{\psi}{\sqrt{M}}\right) + \mathcal{W}\psi, \tag{4.35}$$

where $\mathcal{W} = -\frac{1}{\sqrt{M}}\frac{\psi_0'}{\psi_0}$ is the superpotential, and ψ_0 represents the ground state. The Hamiltonian for a system with an effective mass given in Eq. (4.9) can be factorized as

$$H_{eff} = \mathcal{A}^{\dagger}\mathcal{A} = -\frac{d}{dx}\left(\frac{1}{M(x)}\right)\frac{d}{dx} + V_{eff} \tag{4.36}$$

The corresponding SUSY partner Hamiltonian is given by

$$H^{p}_{eff} = \mathcal{A}\mathcal{A}^{\dagger} = -\frac{d}{dx}\left(\frac{1}{M(x)}\right)\frac{d}{dx} + V^{p}_{eff}. \tag{4.37}$$

We observe that both partner Hamiltonians, H_{eff} and H^{p}_{eff}, describe particles with the same spatial dependence on effective mass but in different potentials, V_{eff} and V^{p}_{eff}, respectively. These SUSY partner potentials are

$$V_{eff} = -\left(\frac{\mathcal{W}}{\sqrt{M}}\right)' + \mathcal{W}^2, \tag{4.38}$$

$$V^{p}_{eff} = V_{eff} + \frac{2\mathcal{W}'}{\sqrt{M}} - \left(\frac{1}{\sqrt{M}}\right)\left(\frac{1}{\sqrt{M}}\right)''. \tag{4.39}$$

If these potentials are shape invariant, they must satisfy the following condition

$$V^{p}_{eff}(x, a_1) = V_{eff}(x, a_2) + R(a_1), \tag{4.40}$$

where $R(a_1)$ is independent of x, a_2 is a function of a_1, and a_1 is a set of parameters. For unbroken SUSY, the eigenvalues and eigenfunctions of the two Hamiltonians with position-dependent effective mass are related as follows for $(n = 0, 1, 2, ...)$

$$E^{eff,p}_{n} = E^{eff}_{n+1}, \qquad E^{eff}_{0} = 0 \tag{4.41}$$

$$\psi^{eff,p}_{n} = [E^{eff}_{n+1}]^{-\frac{1}{2}}\mathcal{A}\psi^{eff}_{n+1}, \tag{4.42}$$

$$\psi^{eff}_{n+1} = [E^{eff,1}_{n}]^{-\frac{1}{2}}\mathcal{A}^{\dagger}\psi^{eff,1}_{n}. \tag{4.43}$$

The superpotentials associated with Cases 1 and 2 of the previous section are given by

$$\mathcal{W}^{I} = \frac{b}{2}\left((1+\alpha)e^{\frac{bx}{2}} - e^{\frac{-bx}{2}}\left(1 + 2\left(\frac{L^{\alpha}_{m-1}(-e^{-bx})}{L^{\alpha-1}_{m}(-e^{-bx})} - \frac{L^{\alpha+1}_{m-1}(-e^{-bx})}{L^{\alpha}_{m}(-e^{-bx})}\right)\right)\right), \tag{4.44}$$

and

$$\mathcal{W}^{II} = \left(\frac{1}{2} - \left(\frac{L^{\alpha+1}_{m-1}(-x^{l})}{L^{\alpha}_{m}(-x^{l})} - \frac{L^{\alpha}_{m-1}(-x^{l})}{L^{\alpha-1}_{m}(-x^{l})}\right)\right)x^{\frac{l}{2}} + \frac{1 - l(\alpha+1)}{2l}x^{-\frac{l}{2}}. \tag{4.45}$$

respectively, here $l = \frac{2-\nu}{1-\nu}$.
The partner potentials corresponding to the superpotential $\mathcal{W}^{I}$ (case 1) can be obtained

using Eqs. (4.38) and (4.39).

$$\begin{aligned} V_{eff}^{I} &= \frac{b^2}{4}\left(e^{bx}\left(\alpha^2-1\right)+e^{-bx}\right)+b^2\left(\frac{2L_{m-1}^{\alpha}(-e^{-bx})^2}{(L_m^{\alpha-1}(-e^{-bx}))^2}-\frac{L_{m-2}^{\alpha+1}(-e^{-bx})}{L_m^{\alpha-1}(-e^{-bx})}\right.\\ &\left.+\frac{L_{m-1}^{\alpha+1}(-e^{-bx})}{\alpha L_m^{\alpha-1}(-e^{-bx})}+\frac{L_{m-1}^{\alpha}(-e^{-bx})}{L_m^{\alpha-1}(-e^{-bx})}-\frac{L_{m-1}^{\alpha+1}(-e^{-bx})}{L_m^{\alpha-1}(-e^{-bx})}\right)e^{-bx}\\ &-b^2\left(\frac{2m+\alpha+\alpha^2}{2\alpha}\right), \end{aligned} \tag{4.46}$$

and

$$\begin{aligned} V_{eff}^{I,p} &= \frac{b^2}{4}\left(e^{bx}\alpha\left(\alpha+2\right)+e^{-bx}\right)+b^2\left(\frac{2L_{m-1}^{\alpha+1}(-e^{-bx})^2}{(L_m^{\alpha}(-e^{-bx}))^2}-\frac{L_{m-2}^{\alpha+2}(-e^{-bx})}{L_m^{\alpha}(-e^{-bx})}\right.\\ &\left.+\frac{L_{m-1}^{\alpha+2}(-e^{-bx})}{(\alpha+1)L_m^{\alpha}(-e^{-bx})}+\frac{L_{m-1}^{\alpha+1}(-e^{-bx})}{L_m^{\alpha}(-e^{-bx})}-\frac{L_{m-1}^{\alpha+2}(-e^{-bx})}{L_m^{\alpha}(-e^{-bx})}\right)e^{-bx}\\ &-b^2\left(\frac{\alpha+1}{2}+\frac{m}{\alpha+2}\right). \end{aligned} \tag{4.47}$$

Similarly, the partner potentials associated with the superpotential $\mathcal{W}^{II}$ (case 2) can be expressed using Eqs. (4.38) and (4.39)

$$\begin{aligned} V_{eff}^{II} &= \frac{1}{4}\left(x^{\frac{\nu-2}{1-\nu}}\left(\alpha^2-1\right)+x^{\frac{2-\nu}{1-\nu}}\right)+\left(\frac{2L_{m-1}^{\alpha}(-x^{\frac{2-\nu}{1-\nu}})^2}{(L_m^{\alpha-1}(-x^{\frac{2-\nu}{1-\nu}}))^2}-\frac{L_{m-2}^{\alpha+1}(-x^{\frac{2-\nu}{1-\nu}})}{L_m^{\alpha-1}(-x^{\frac{2-\nu}{1-\nu}})}\right.\\ &\left.+\frac{L_{m-1}^{\alpha+1}(-x^{\frac{2-\nu}{1-\nu}})}{\alpha L_m^{\alpha-1}(-x^{\frac{2-\nu}{1-\nu}})}+\frac{L_{m-1}^{\alpha}(-x^{\frac{2-\nu}{1-\nu}})}{L_m^{\alpha-1}(-x^{\frac{2-\nu}{1-\nu}})}-\frac{L_{m-1}^{\alpha+1}(-x^{\frac{2-\nu}{1-\nu}})}{L_m^{\alpha-1}(-x^{\frac{2-\nu}{1-\nu}})}\right)x^{\frac{2-\nu}{1-\nu}}\\ &+\frac{(1-\nu)(3-\nu)}{4(2-\nu)^2}x^{\frac{\nu-2}{1-\nu}}-\frac{2m+\alpha+\alpha^2}{2\alpha}, \end{aligned} \tag{4.48}$$

and

$$\begin{aligned} V_{eff}^{II,p} &= \frac{1}{4}\left(x^{\frac{\nu-2}{1-\nu}}\alpha(\alpha+2)+x^{\frac{2-\nu}{1-\nu}}\right)+\left(\frac{2L_{m-1}^{(\alpha+1)}(-x^{\frac{2-\nu}{1-\nu}})^2}{(L_m^{\alpha}(-x^{\frac{2-\nu}{1-\nu}}))^2}-\frac{L_{m-2}^{(\alpha+2)}(-x^{\frac{2-\nu}{1-\nu}})}{L_m^{\alpha}(-x^{\frac{2-\nu}{1-\nu}})}\right.\\ &\left.+\frac{L_{m-1}^{(\alpha+2)}(-x^{\frac{2-\nu}{1-\nu}})}{(\alpha+1)L_m^{\alpha}(-x^{\frac{2-\nu}{1-\nu}})}+\frac{L_{m-1}^{(\alpha+1)}(-x^{\frac{2-\nu}{1-\nu}})}{L_m^{\alpha}(-x^{\frac{2-\nu}{1-\nu}})}-\frac{L_{m-1}^{(\alpha+2)}(-x^{\frac{2-\nu}{1-\nu}})}{L_m^{\alpha}(-x^{\frac{2-\nu}{1-\nu}})}\right)x^{\frac{2-\nu}{1-\nu}}\\ &+\frac{(1-\nu)(3-\nu)}{4(2-\nu)^2}x^{\frac{\nu-2}{1-\nu}}-\left(\frac{\alpha}{2}+\frac{m}{\alpha+1}\right). \end{aligned} \tag{4.49}$$

The effective potentials V_{eff}^{I} and V_{eff}^{II}, derived in Eq. (4.46) and (4.48) respectively, are

identical to the potentials given in Eq. (4.23) and (4.32) when $V_c^I = -b^2\left(\frac{2m+\alpha+\alpha^2}{2\alpha}\right)$ and $V_c^{II} = -\frac{2m+\alpha+\alpha^2}{2\alpha}$.
By examining Eqs. (4.46) and (4.47), we observe that the potential and its SUSY counterpart fulfill the following condition

$$V_{eff}^{I,p}(x,\alpha) = V_{eff}^{I}(x,\alpha+1) + b^2. \tag{4.50}$$

Similarly, from Eqs. (4.48) and (4.49), it is evident that the partner potentials satisfy a similar relation.

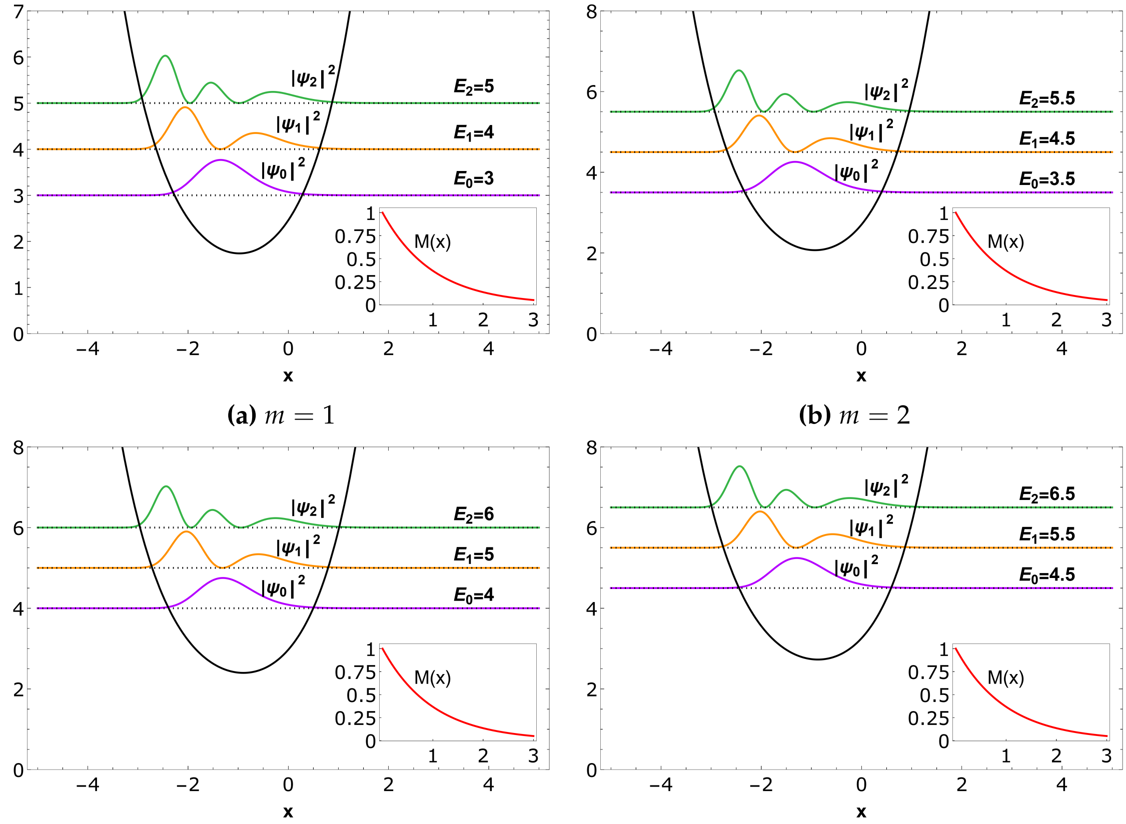
$$V_{eff}^{II,p}(x,\alpha) = V_{eff}^{II}(x,\alpha+1) + 1. \tag{4.51}$$

The potentials derived for case 1 and case 2 clearly satisfy the condition of SI as defined in Eq. (4.40). Therefore, we conclude that the newly introduced potentials demonstrate SI.
We determine the eigenstates for the partner potential corresponding to the potential (4.23) as

$$\psi_{n,m}^{p}(x) \propto \frac{\exp\left[-\frac{1}{2}((\alpha+2)bx + e^{-bx})\right]}{L_m^{\alpha}(-e^{-bx})}\hat{L}_{n+m,m}^{\alpha+1}(e^{-bx}) \quad n = 0,1,2,... \tag{4.52}$$

and the eigenstate of the partner potential of Eq. (4.32) is given by

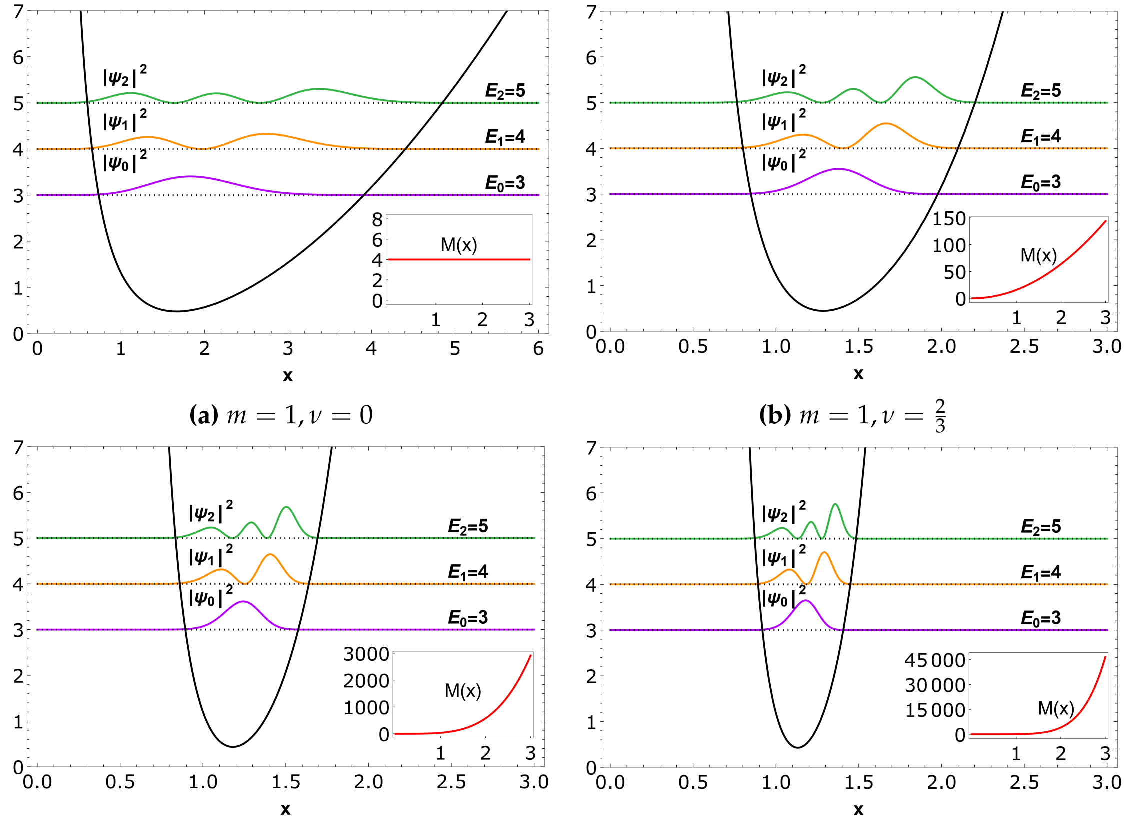
$$\psi_{n,m}^{p}(x) \propto \frac{\exp\left(\frac{-x^{\frac{2-\nu}{1-\nu}}}{2}\right)}{\left(L_m^{\alpha}(-x^{\frac{2-\nu}{1-\nu}})\right)} x^{\frac{(2+\alpha)+(\alpha+1)(1-\nu)}{2(1-\nu)}}\hat{L}_{n+m,m}^{\alpha+1}(x^{\frac{2-\nu}{1-\nu}}), \quad n = 0,1,2,... \tag{4.53}$$



(a) $m = 1$ **(b)** $m = 2$

(c) $m = 3$ **(d)** $m = 4$

Figure 4.6: Plot of the supersymmetric partner potential $V_{eff}^{I,p}$, in Eq. (4.47), for various m values, as well as the square of the first three bound state wavefunctions $|\psi_0|^2$ (purple line), $|\psi_1|^2$ (orange line), and $|\psi_2|^2$ (green line), correspond to distinct m values for the mass function $M(x)$ (red line) as defined in Eq. (4.22). We have choosen $\alpha = 2$, $b = 1$.



(a) $m = 1, \nu = 0$ **(b)** $m = 1, \nu = \frac{2}{3}$

(c) $m = 1, \nu = \frac{4}{5}$ **(d)** $m = 1, \nu = \frac{6}{7}$

Figure 4.7: Plot of the partner potential $V_{eff}^{II,p}$ given in Eq. (4.49) for $m = 1$ and different ν values, square of first three associated bound state wavefunctions $|\psi_0|^2$ (purple line), $|\psi_1|^2$ (orange line), $|\psi_2|^2$ (green line) corresponding to different ν values and mass function $M(x)$ (red line) defined in case 2. We have considered here $\alpha = 2$.

4.4 Results and Discussions

The investigation of systems with PDM holds significant interest across various branches of physics due to its crucial relevance in diverse physical scenarios. However, exact solutions are only available for a limited number of PDM systems with specific forms of PDM terms, such as $M(x) \propto [g'(x)]^{\mu}$, where $\mu = 1, 2$, and -1. In this chapter, we have explored a novel form of PDM term proportional to $(g'(x))^{\nu}$ with $\nu = \frac{2\eta}{2\eta+1}$, where $\eta = 0, 1, 2, \ldots$. We derived solutions for the Schrödinger equation with PDM in terms ofX_m-Laguerre polynomials. Additionally, we have derived a one-parameter family of isochronous potentials that are ES. The solutions corresponding to these potentials are demonstrated to be expressed in terms of X_m-Laguerre polynomials. We have demonstrated SUSY for PDM systems, and for this particular system, we have identified a one-parameter family of isochronous potentials along with their partner potentials. We demonstrate that the exact solvability of the system arises from the underlying SI of these SUSY potentials.